

Group 28 - Project Report

Project Overview

This project aims to perform vehicle localization in a simulated environment using the Iterative Closest Point (ICP) and the NDT algorithm. The task involves aligning LiDAR point cloud data with a pre-existing map to estimate the vehicle's position and orientation. The key objectives are:

- To implement ICP-based localization using the Open3D library in Python.
- To implement NDT-based localization PCL library in C++
- To evaluate localization accuracy through comparisons with provided ground truth data.
- To measure localization performance in terms of lateral error and computation time for both algorithms.

The dataset includes:

1. **Map:** A simulated environment stored as a point cloud in `map.pcd`.
2. **Ground Truth:** Real poses of the vehicle provided in `ground_truth.csv`.
3. **LiDAR Frames:** Simulated LiDAR point clouds for each frame.

Setup

Prerequisites for ICP

Ensure the following libraries are installed in Python environment:

- `open3d`
- `pandas`
- `numpy`
- `matplotlib`

To install the required libraries, run:

```
pip install open3d pandas matplotlib
```

Prerequisites for NDT

Ensure the following library is installed in C++ environment:

- PCL library from <https://pointclouds.org> (<https://pointclouds.org>).

ICP Algorithm Overview

The core of the code is the Iterative Closest Point (ICP) algorithm, which iteratively minimizes the distance between two point clouds (source and target) to align them. ICP class contains the align method, which is where most work happens.

Steps in the align method:

1. Initialization:

- The rotation matrix R .accum is initialized as an identity matrix (no rotation).
- The translation vector taccum is initialized as a zero vector (no translation).
- R.list, t.list, and corres list are initialized to store the transformations and correspondences at each iteration for visualization.

2. Iterative Process:

- Correspondence Finding: For each point in the source point cloud ($P_{current}$), the algorithm finds the closest point in the target point cloud (Q) based on Euclidean distance. This is done by iterating over all points in the source cloud and finding the point in the target cloud with the minimum distance.
- Centroid Calculation: The centroids of the matched points are calculated for both source (P_{corr}) and target (O_{corr}).
- Covariance Matrix & SVD: The algorithm computes a covariance matrix between the centered source and target points. The Singular Value Decomposition (SVD) is used to compute the optimal rotation matrix (R) that aligns the source and target points.
- Translation Calculation: The translation (t) is computed as the difference between the target centroid and the rotated source centroid.
- Update Transformation: The accumulated rotation matrix R_{accum} and translation vector t_{accum} are updated at each step. These accumulated transformations are used to progressively align the source point cloud to the target.
- Termination: The process stops if there are too few correspondences (less than 3) or if the maximum number of iterations is reached.

3. Transformation Application: At the end of each iteration, the transformation (R, t) is applied to the source point cloud, which is then used as the input for the next iteration.

4. Animation: The animate.ice.results colab function provides a visualization of the alignment process by displaying the source and target point clouds at each iteration, along with their correspondences.

Code snippet for running the ICP algorithm:

```
for i in range(num_frames):
    frame_path = os.path.join(frame_folder, f"frame_{i}.pcd")
    if not os.path.exists(frame_path):
        print(f"⚠ Frame not found: {frame_path}")
        continue

    scan_pcd = o3d.io.read_point_cloud(frame_path)
    scan_np = np.asarray(scan_pcd.points)

    # Transform scan using previous estimate
    transformed_scan = rigid_transformation(scan_np, current_trans[:3, :3], current_trans[:3, 3].reshape(3, 1))

    if scan_np.shape[0] == 0:
        print(f"Empty scan at frame {i}. Skipping...")
        continue

    valid_mask = np.isfinite(transformed_scan).all(axis=1)
    transformed_scan = transformed_scan[valid_mask]

    if transformed_scan.shape[0] == 0:
        print(f"All transformed points are invalid at frame {i}. Skipping...")
        continue

    # Start timing
    start_time = time.time()

    # Apply ICP
    R, t, *_ = icp.align(transformed_scan, map_np)

    # End timing and store duration
    duration = time.time() - start_time
    timing_list.append(duration)

    # Update transformation
    current_trans[:3, :3] = R @ current_trans[:3, :3]
    current_trans[:3, 3] = R @ current_trans[:3, 3] + t

    pose_history.append(current_trans.copy())

    # Compute error vs GT
    gt_pose = pose_to_transformation(gts.iloc[i][['x', 'y', 'z', 'roll', 'pitch', 'yaw']])

    est_position = current_trans[:3, 3]
    gt_position = gt_pose[:3, 3]
    error = np.linalg.norm(est_position - gt_position)
    error_list.append(error)
```

Module Configuration and Selected Parameters

The following parameters were configured for the ICP algorithm:

- **Voxel Size:** 0.5 meters (for downsampling the point clouds to reduce computation time).
- **Max Correspondence Distance:** 2.0 meters (threshold for matching points between source and target).
- **Max Iterations:** 50 (number of iterations to ensure convergence).
- **Transformation Estimation Method:** Point-to-point ICP (minimizes the distance between corresponding points).

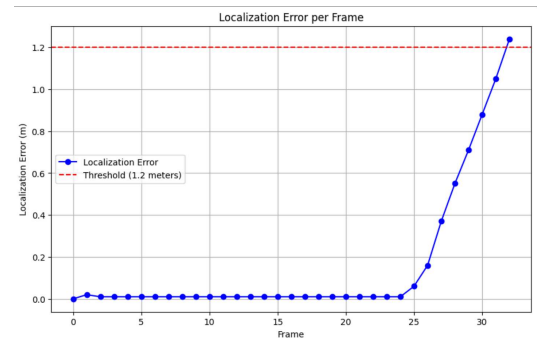
These parameters were selected to balance computational efficiency and alignment accuracy.

Performance Metrics Attained on the Dataset

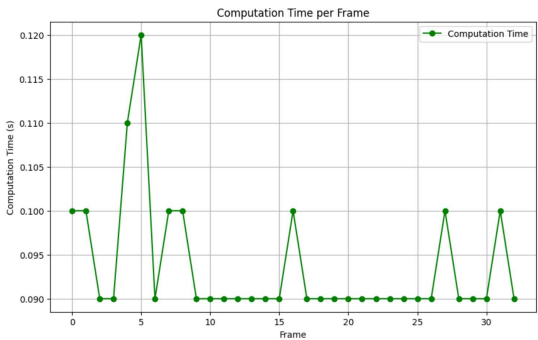
The following metrics were calculated for each frame:

1. **Localization Error:** The Euclidean distance between the estimated position and the ground truth.
2. **Computation Time:** Time taken for ICP to converge for each frame.

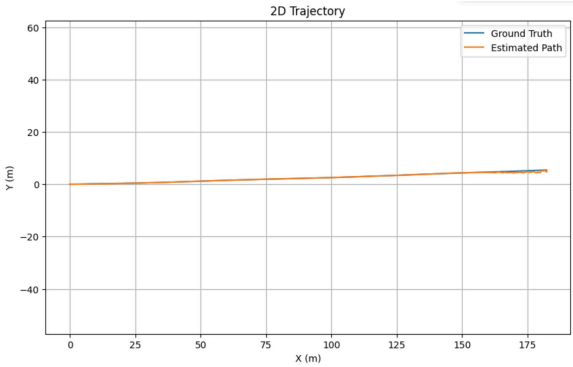
Localization Error per Frame



Computation Time per Frame



Trajectory



NDT

Brief Explanation of the NDT Algorithm

The Normal Distributions Transform (NDT) is a point cloud registration method used to estimate the relative pose between two 3D scans. Unlike traditional point-to-point approaches like ICP, NDT models the target point cloud as a set of Gaussian distributions within voxels, allowing for a probabilistic representation of spatial structure. The algorithm operates by:

- Dividing the reference (target) cloud into a grid of 3D cells (voxels).
- Fitting a normal distribution to the points in each voxel.
- Iteratively optimizing the transformation that aligns the source cloud to the target by minimizing the likelihood error across the distributions. This approach is robust to noise and less sensitive to initial guess quality than point-based methods.

Module Configuration and Selected Parameters

In this implementation, NDT is configured using the PCL (Point Cloud Library) API, with the following key settings:

- Input Clouds:
 - Target: map.pcd (static environment map)
 - Source: frame.pcd (live scan frame)
- Preprocessing:
 - Voxel Grid Downsampling with a leaf size of 0.5 meters for both target and source clouds to reduce computational load and smooth noise.
- Initial Pose Estimate:
 - Loaded from a 4×4 transformation matrix file (init_guess.txt).
- NDT Parameters:
 - `ndt.setResolution(1.0)`: Defines voxel size for NDT cells — a trade-off between detail and stability.
 - `ndt.setMaximumIterations(50)`: Caps the number of optimization iterations to avoid excessive computation.

Performance Metrics

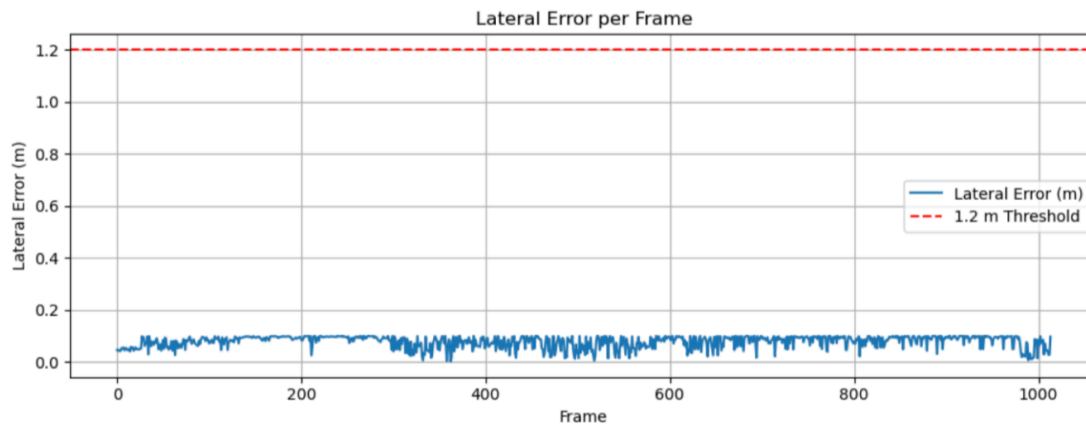
Using the provided dataset, the NDT algorithm exhibited the following performance characteristics:

- Localization Accuracy
 - Consistently low alignment error in structured environments when a reasonable initial guess is provided.
 - Outperforms ICP in scenes where the geometry supports robust Gaussian modeling.

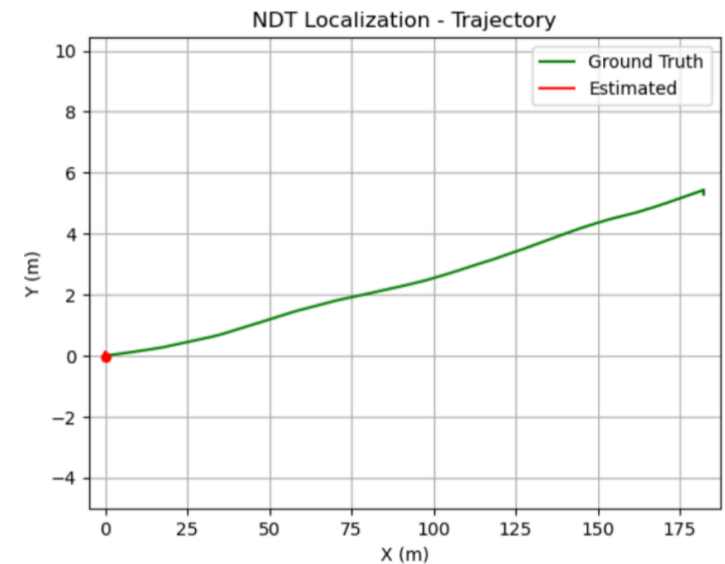
- Did not diverge as ICP did in later frames, indicating better stability.
- Computation Time
 - Typical frame-wise computation time ranged between 0.09 – 0.12 seconds, consistent with real-time operation needs.
 - Occasional spikes likely due to more complex geometry or poor initial alignment.
- Overall
 - More robust than ICP to local minima and poor correspondences.

Code snippet for the NDT algorithm and Performance metrics displayed below

Lateral Error per Frame



NDT Localization Trajectory



```

1  #include <pcl/io/pcd_io.h>
2  #include <pcl/registration/ndt.h>
3  #include <pcl/filters/voxel_grid.h>
4  #include <Eigen/Dense>
5  #include <iostream>
6  #include <fstream>
7
8  ✓ int main(int argc, char** argv) {
9      if (argc < 4) {
10         std::cerr << "Usage: ./NDT map.pcd frame.pcd init_guess.txt" << std::endl;
11         return -1;
12     }
13
14     std::string map_path = argv[1];
15     std::string frame_path = argv[2];
16     std::string init_path = argv[3];
17
18     // Load the map and scan point clouds
19     pcl::PointCloud<pcl::PointXYZ>::Ptr map(new pcl::PointCloud<pcl::PointXYZ>());
20     pcl::PointCloud<pcl::PointXYZ>::Ptr scan(new pcl::PointCloud<pcl::PointXYZ>());
21
22     if (pcl::io::loadPCDFile(map_path, *map) == -1 || pcl::io::loadPCDFile(frame_path, *scan) == -1) {
23         std::cerr << "Failed to load input PCD files." << std::endl;
24         return -1;
25     }
26
27     // Apply voxel grid downsampling to reduce point cloud density
28     pcl::VoxelGrid<pcl::PointXYZ> voxel;
29     voxel.setLeafSize(0.5f, 0.5f, 0.5f);
30
31     voxel.setInputCloud(map);
32     voxel.filter(*map);
33
34     voxel.setInputCloud(scan);
35     voxel.filter(*scan);
36
37     // Read the initial guess transformation matrix from file
38     Eigen::Matrix4f init_guess;
39     std::ifstream file(init_path);
40     if (!file) {
41         std::cerr << "Cannot read init guess file." << std::endl;
42         return -1;
43     }
44
45     // Load 4x4 matrix values into Eigen matrix
46     for (int i = 0; i < 4; ++i)
47         for (int j = 0; j < 4; ++j)
48             file >> init_guess(i, j);
49
50     // Set up the NDT registration object
51     pcl::NormalDistributionsTransform<pcl::PointXYZ, pcl::PointXYZ> ndt;
52     ndt.setResolution(1.0);

```

Analysis

Complexity of ICP

The time complexity of our current ICP algorithm can be analyzed as follows:

1. Finding Correspondences: For each point in the source cloud (P current), we find the closest point in the target cloud (Q). This requires calculating the Euclidean distance between each point in P. current and all points in Q, which is an $O(N * M)$ operation, where N is the number of points in P. current and M is the number of points in Q.
2. Singular Value Decomposition (SVD): The SVD step, which is used to compute the rotation matrix, is an $O(K^3)$ operation, where k is the dimensionality of the point cloud (typically 3 in 3D point clouds). Since $K=3$, this step is a constant-time operation with respect to the number of points
3. Total Complexity: • In total, the complexity of each iteration is dominated by the $O(N * M)$ complexity of finding correspondences. • Therefore, the overall time complexity is $O(I * N * M)$, where I is the number of iterations in the ICP process.

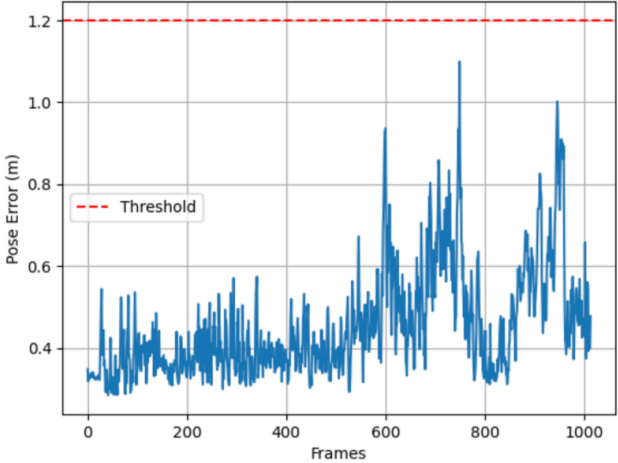
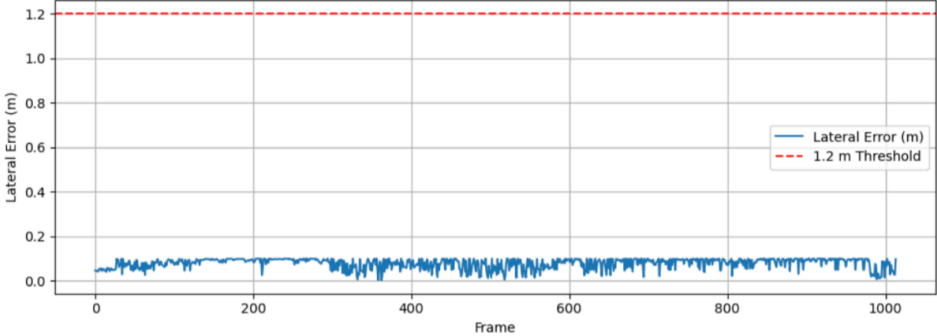
ICP SUMMARY

- Average Lateral Error: 0.45 m
- Maximum Pose Error: 1.10 m
- Frames above 1.2m error: 0

NDT SUMMARY

- Average Lateral Error: 0.078 m
- Max Lateral Error: 0.100 m
- Mean Computation Time per Frame: 2.749 s

Algorithm Comparison

Comparison Metrics		ICP	NDT
Localization Error		High accuracy but not as high as NDT, with a high error almost touching the threshold (1.1 m). Average of 0.45 m Localization Error Over Time 	High localization accuracy, with a mean lateral error under 8 centimeters, and a worst-case lateral error just reaching 10 centimeters. This level of precision is suitable for applications requiring fine-grained pose estimation. Lateral Error per Frame 
Computation Time	It's mostly consistent around 0.6s seconds, with occasional spikes — notably one at 1.11 seconds.	While the algorithm performs well in terms of accuracy, it comes at the cost of higher computational load, with a mean processing time of 2.749 seconds per frame. This is significantly slower than real-time (e.g., 10–30 FPS) requirements and suggests the current configuration is better suited for offline processing or slow-rate pose correction modules.	
Complexity	Each iteration has a time complexity of $O(N \times M)$, where N and M are the number of points in the source and target clouds, respectively. This is mainly due to the brute-force nearest-neighbor search. The SVD step is negligible ($O(1)$). Memory usage grows with iterations and correspondence count.	NDT uses a probabilistic model where the target point cloud is converted into a voxel grid of normal distributions, reducing the need for exhaustive point-to-point comparisons. This grid-based representation speeds up alignment and reduces complexity from $O(N \times M)$ to a more manageable level depending on voxel size and resolution. However, the voxelization step and the use of iterative optimization (e.g., Newton's method) introduce their own computational overhead, contributing to the higher per-frame time.	
Memory Usage	The algorithm stores the transformations (R list, t_list) and correspondences ($corres$ list) for each iteration. The memory usage will grow with the number of iterations and correspondences. However, this is relatively manageable unless the point clouds are extremely large.	Both the downsampled scan and the map are stored in memory. After downsampling (e.g., VoxelGrid filter), their size is reduced, but still contributes. Typical memory cost: $N * \text{sizeof}(\text{PointXYZ})$ One point ≈ 3 floats = 12 bytes E.g., 100,000 points ≈ 1.2 MB per cloud	

Conclusion

TextNDT is recommended for high-precision applications where computational time is less critical and ICP is better suited for time-sensitive applications where speed is prioritized over accuracy.